

Spline-SVP v1: Weekly Progress Report

1 Objectives and progress

The objective of this week was to obtain an implementable first version of Smallest Valid Partitioning for continuous piecewise-linear signals. The current formulation is called *spline-SVP v1*.

The main progress was:

- (i) fixing the mathematical definition of the estimator;
- (ii) deriving and implementing the continuous piecewise-linear fitting cost;
- (iii) implementing and testing a first state-independent validity statistic;
- (iv) implementing an exhaustive reference oracle and checking it on a small synthetic example.

The exhaustive implementation is intended as a correctness reference for small n , not as the final efficient algorithm.

2 Spline representation and fitting cost

We consider

$$y_i = f_0(x_i) + \varepsilon_i, \quad i = 1, \dots, n,$$

with

$$x_1 < \dots < x_n, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

where f_0 is continuous and piecewise linear. For the first implementation, σ^2 is treated as known.

A candidate partition is represented by

$$\tau = (\tau_0, \dots, \tau_K), \quad 1 = \tau_0 < \tau_1 < \dots < \tau_K = n.$$

The partition contains K linear pieces and $K - 1$ interior changepoints.

For a fixed partition, define the fitted boundary states

$$z_k = f(x_{\tau_k}), \quad k = 0, \dots, K.$$

For a generic segment $(s, t]$, with endpoint states

$$u = f(x_s), \quad v = f(x_t),$$

the fitted affine function is

$$\ell_{s,t}^{u,v}(x) = u + \frac{v - u}{x_t - x_s}(x - x_s).$$

The segment fitting cost is

$$C_{s,t}(u, v) = \sum_{i=s+1}^t [y_i - \ell_{s,t}^{u,v}(x_i)]^2.$$

For a complete partition,

$$C(\tau, z) = (y_1 - z_0)^2 + \sum_{k=1}^K C_{\tau_{k-1}, \tau_k}(z_{k-1}, z_k).$$

The same state z_k is used by the two adjacent segments at x_{τ_k} . Therefore

$$\ell_{\tau_{k-1}, \tau_k}^{z_{k-1}, z_k}(x_{\tau_k}) = z_k = \ell_{\tau_k, \tau_{k+1}}^{z_k, z_{k+1}}(x_{\tau_k}),$$

so continuity is enforced structurally.

For a fixed partition, define

$$z^*(\tau) = \arg \min_z C(\tau, z),$$

and

$$C^*(\tau) = C(\tau, z^*(\tau)).$$

The problem can be written as

$$C(\tau, z) = \|y - X_\tau z\|_2^2,$$

so the optimal boundary states are obtained by standard linear least squares. In the R implementation, this is solved using `lm.fit()`.

3 State-independent validity statistic

The first validity criterion does not depend on the endpoint states (u, v) . Instead, it asks whether the observations on a candidate interval are sufficiently well described by one affine function, relative to an alternative allowing one internal slope change.

For an interval $(s, t]$, define

$$RSS_0(s, t) = \min_{a, b} \sum_{i=s+1}^t (y_i - a - bx_i)^2.$$

For a candidate internal knot r , define the continuous one-knot model

$$g_{a, b, c, r}(x) = a + bx + c(x - x_r)_+,$$

with

$$RSS_1(s, r, t) = \min_{a, b, c} \sum_{i=s+1}^t [y_i - a - bx_i - c(x_i - x_r)_+]^2.$$

The current validity statistic is

$$T_{s, t} = \frac{1}{\sigma^2} \left[RSS_0(s, t) - \min_{s < r < t} RSS_1(s, r, t) \right].$$

The interval is valid when

$$T_{s, t} \leq \gamma.$$

Since the one-knot alternative contains the affine model as the case $c = 0$,

$$RSS_1(s, r, t) \leq RSS_0(s, t),$$

and therefore $T_{s, t} \geq 0$, up to numerical precision.

For intervals with at most two observations, i.e.

$$t - s \leq 2,$$

the affine model can interpolate all observations exactly, so

$$RSS_0(s, t) = 0$$

and consequently

$$T_{s,t} = 0.$$

This value is used directly in the implementation.

The normalization by σ^2 puts the reduction in RSS on the Gaussian likelihood-ratio scale. It also gives invariance under a common rescaling of both y and σ .

Several basic tests were performed. For a perfectly affine noiseless signal, the implementation gave

$$T = 3.23 \times 10^{-29},$$

which is zero up to floating-point precision. For a clear piecewise-linear signal with one slope change, it gave

$$T = 30.$$

4 Spline-SVP v1 estimator and exhaustive oracle

Let

$$\mathcal{V}_\gamma = \{\tau : T_{\tau_{k-1}, \tau_k} \leq \gamma \text{ for all } k\}.$$

The minimum number of segments among valid partitions is

$$K^* = \min_{\tau \in \mathcal{V}_\gamma} K(\tau).$$

Among partitions attaining K^* , the estimator selects

$$\hat{\tau} \in \arg \min_{\substack{\tau \in \mathcal{V}_\gamma \\ K(\tau) = K^*}} C^*(\tau).$$

Thus the optimization is lexicographic: the number of segments is minimized first, and the fitting cost is used only to break ties among valid partitions with the same minimum number of segments.

After selecting $\hat{\tau}$, define

$$\hat{z} = z^*(\hat{\tau}),$$

and construct the fitted signal by affine interpolation between the estimated boundary states.

The first reference algorithm enumerates all candidate partitions. There are exactly

$$2^{n-2}$$

partitions because each index $2, \dots, n-1$ is either included or excluded as an interior boundary.

For every partition, the algorithm:

1. checks validity of all its segments;
2. rejects the partition if any segment is invalid;
3. solves the continuous least-squares problem for the boundary states;
4. records $K(\tau)$ and $C^*(\tau)$.

The oracle then selects the lexicographically optimal valid partition. Since every candidate partition is checked, this provides a global solution of the currently defined v1 estimator for small n . Its complexity is exponential in n .

5 First synthetic example

The implementation was tested on

$$n = 10, \quad x_i = i, \quad \sigma = 0.25,$$

with true signal

$$f_0(x) = \begin{cases} 0.5x, & x \leq 5, \\ 2.5 - 0.7(x - 5), & x > 5. \end{cases}$$

Gaussian noise was added independently.

For this first implementation check, the threshold was set arbitrarily to

$$\gamma = 4.$$

This value was used only for testing and has not yet been calibrated.

For $n = 10$, the oracle enumerated

$$2^8 = 256$$

partitions, of which 222 were valid.

The selected partition was

$$\hat{\tau} = (1, 5, 10),$$

so that

$$K^* = 2.$$

The estimated boundary states were

$$\hat{z} = (0.3953865, 2.6107543, -0.9687846),$$

and the optimal fitting cost was

$$C^*(\hat{\tau}) = 0.3022311.$$

The validity statistics for the two selected segments were

$$T_{1,5} = 0.5851 < 4, \quad T_{5,10} = 1.4276 < 4,$$

whereas for the complete one-segment interval,

$$T_{1,10} = 82.5021 > 4.$$

Therefore a one-segment solution is invalid, while the selected two-segment partition is valid.

Figure 1 shows the result.

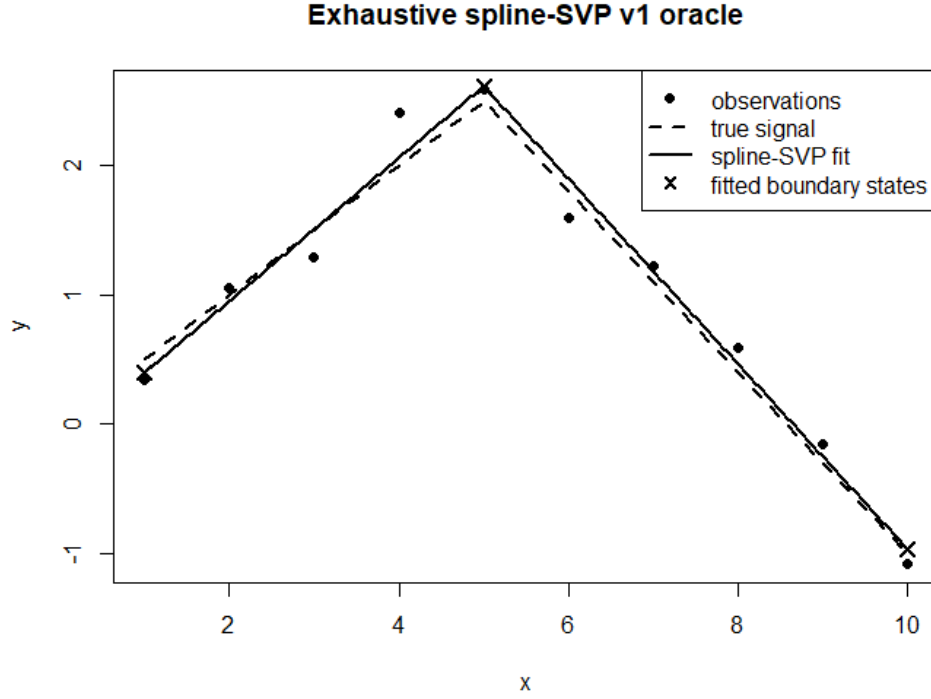


Figure 1: First synthetic test of the exhaustive spline-SVP v1 oracle with $n = 10$, $\sigma = 0.25$, and test value $\gamma = 4$. The fitted signal is continuous and the estimated changepoint is at index 5.

This example is only an implementation check. The successful recovery of the simulated changepoint from one realization should not be interpreted as a general statistical performance result.